Student Name: Shiva Sai Krishna S | **Register Number:** 1911260318

Course: CSA0421 - Operating System | **GitHub Repo:** krishshiva721/CSA04--Assignment1

TASK 8: CPU STATES AND HARDWARE PROTECTION

Dual-Mode Hardware Architecture, Interrupt Mechanisms, and Memory Isolation

1. DUAL-MODE OPERATION & PRIVILEGED INSTRUCTIONS

Modern CPUs enforce structural isolation via a bit in the processor status register (**Mode Bit**: 0 for Kernel/Supervisor Mode, 1 for User Mode). Instructions classified as *Privileged* (such as `cli` to disable interrupts, modifying page base registers `CR3`, or direct hardware port I/O) can only execute when CPU is in Kernel Mode.

2. CATASTROPHIC FAILURE PREVENTION IN BEDSIDE MONITORS

If an unprivileged user application or third-party charting module on a patient monitor were permitted to execute `cli` (Clear Interrupt Flag):

- The CPU would ignore hardware interrupts generated by ECG hardware timers and cardiac sensors.
- If a cardiac arrhythmia event occurred while interrupts were disabled, the CPU would fail to context-switch to the alarm service thread.
- **Hardware Protection Rule:** If a user mode program attempts to execute a privileged instruction, the CPU triggers a *General Protection Fault Trap (Vector 13)*, instantly terminating the faulty user application while keeping kernel alarms active.

3. ALTERNATIVE CONCEPT: INTERRUPT-DRIVEN DETECTION VS. POLLING ANALYSIS

Alternative Scenario: Polling Sensor Loop vs. Hardware Interrupt Architecture

Architectural Metric	Polling Loop Mechanism	Interrupt-Driven Mechanism
Worst-Case Alarm Latency	$T_{delay} = T_{poll_interval}$ (e.g., 50 ms delay if polling at 20 Hz, failing requirements).	$T_{delay} = T_{interrupt_latency}$ (< 5 µs deterministic response).
CPU Utilization & Waste	100% CPU spinning in tight loop checking registers, wasting compute resources.	0% idle CPU footprint; processor stays in low-power state until hardware signal arrives.
Power & Thermal Impact	High energy draw, causing rapid battery drain on portable monitoring devices.	Optimal energy conservation; processor enters sleep states between sensor pulses.